

Beekin Rent Model

Project Summary — Experiments • Refresh Automation • Monitoring & Rollback

Objective	Evaluate census signal + build production-style model lifecycle.
Outcome	Census features added complexity without improving MdAPE; baseline retained.
Metric	MdAPE (Median Absolute % Error) for robust, stakeholder-friendly evaluation.

1) Objective

This work evaluates whether census-based demographic variables add meaningful predictive signal to a baseline rent model, and demonstrates a production-aligned workflow for safe model refresh and governance.

2) Data Used

Historical training data (train.csv) and a hold-out evaluation set (test.csv) were used for training and benchmarking. New market data (data_2022.csv) was used to simulate warehouse ingestion and trigger retraining. Census data (census.csv) was included only for controlled experiments.

3) Experimentation (Baseline vs Census)

A baseline XGBoost model was trained on stable property/location features. A census-enhanced model used the same algorithm and hyperparameters to ensure a fair comparison. The enhanced version did not improve MdAPE and increased operational and fairness risk; therefore it was not recommended for deployment.

4) Model Refresh / Retraining Workflow

A retraining notebook was implemented to refresh the model when new data becomes available. The workflow appends incremental data to historical records, applies the same feature engineering and encoding, retrains the model, and stores versioned model artifacts. Encoding artifacts are saved to ensure consistent preprocessing across refresh cycles.

5) Monitoring & Rollback Governance

A monitoring function compares old vs refreshed model performance on the same evaluation dataset. MdAPE deltas are measured and checked against a tolerance threshold to detect silent degradation. If the refreshed model degrades beyond tolerance, a rollback recommendation is generated to protect production reliability.

6) Product & Risk Considerations

A 1-page memo documents feature risks (bias/explainability), recommended refresh cadence, failure tests for drift and silent degradation, and monotonicity trade-offs to improve trust and user experience.

Overall: This project demonstrates an end-to-end ML workflow that prioritizes reproducibility, safe refresh, monitoring, and rollback controls — aligning model quality with product reliability.